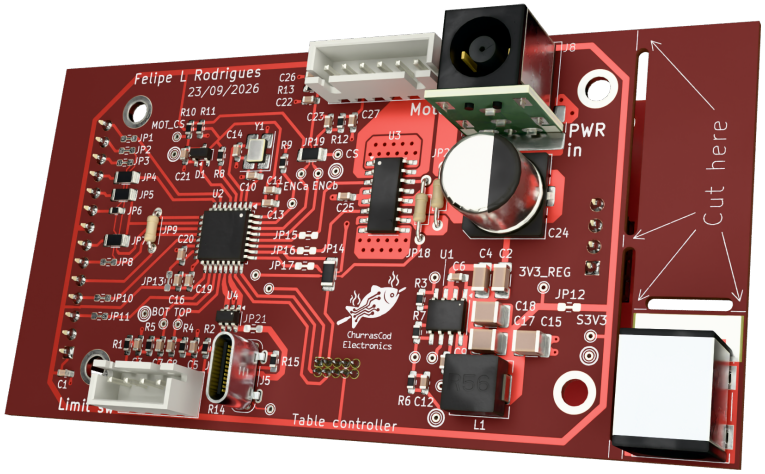
# Desk controller assembly drawing

## Top view




## Bottom view

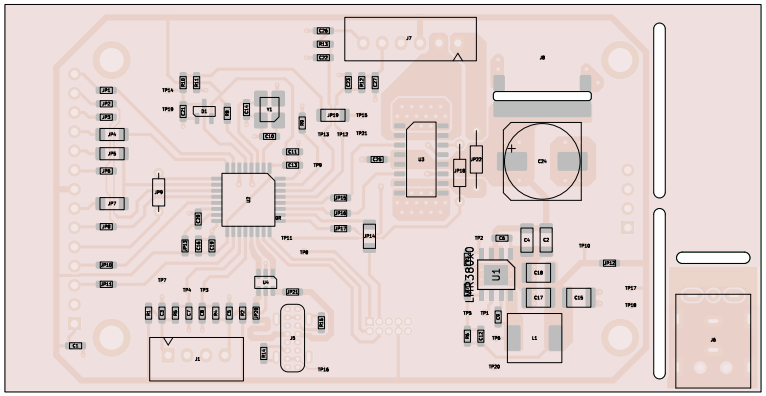


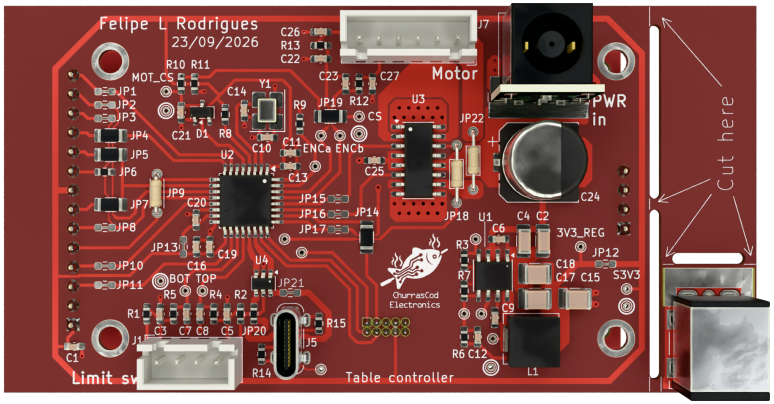

[Interactive BOM]

| | | |
|---|---|---|
| ChurrasCod Electronics | **Desk controller** | Sheet: Assembly overview |
| | **Notes** | File: Desk_controller.kicad_pcb |
| | | Rev: 1.0.1 / Drwn: FLR |
| | | Mat: Concept / Rev'd: FLR |
| | | Date: 06/10/2026 / Rls'd: FLR |
| | | Dwg: NA / Size: A4 |
| | | Generated with KiCad / Page: 1/7 |

# Top assembly (Scale 1:1)

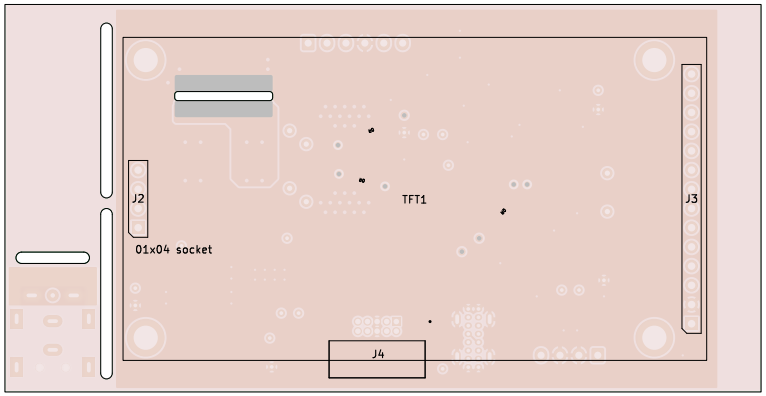
| Type | Fitted | No fit |
|---|---|---|
| SMD | 59 | 0 |
| THT | 6 | 0 |
| Total | 65 | 0 |






| | | |
|---|---|---|
| ChurrasCod Electronics | **Desk controller** | Sheet: Top assembly (Scale 1:1) |
| | **Notes** | File: Desk_controller.kicad_pcb |
| | | Rev: 1.0.1 / Drwn: FLR |
| | | Mat: Concept / Rev'd: FLR |
| | | Date: 06/10/2026 / Rls'd: FLR |
| | | Dwg: NA / Size: A4 |
| | | Generated with KiCad / Page: 2/7 |

# Bottom assembly (Scale 1:1)







| Type | Fitted | Not fitted |
|---|---|---|
| SMD | 3 | 0 |
| THT | 4 | 0 |
| Total | 7 | 0 |

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**Desk controller**

**Notes**

Sheet: Bottom assembly (Scale 1:1)
File: Desk_controller.kicad_pcb

| | |
|---|---|
| Rev: 1.0.1 | Drwn: FLR |
| Mat: Concept | Rev'd: FLR |
| Date: 06/10/2026 | Rls'd: FLR |
| Dwg: NA | Size: A4 |
| Generated with KiCad | Page: 3/7 |

# Top BOM - fitted (1 of 3)

| References | Qty | Value | Footprint | MPN | Manufacturer |
|---|---|---|---|---|---|
| C1 C6 C9 C13 C19 C20 C25 | 7 | 100n | C_0603_1608Metric_Pad1.08x0.95mm_HandSolder | AC0603KRX7R9BB104 | Yageo |
| C3 C5 C26 C27 | 4 | 10n | C_0603_1608Metric_Pad1.08x0.95mm_HandSolder | AC0603KRX7R9BB103 | Yageo |
| C7 C8 C21 C22 C23 | 5 | 1n | C_0603_1608Metric_Pad1.08x0.95mm_HandSolder | AC0603KRX7R9BB102 | Yageo |
| C11 C16 | 2 | 1u | C_0603_1608Metric_Pad1.08x0.95mm_HandSolder | AC0603KRX7R7BB105 | Yageo |
| C24 | 1 | 220u | CP_Elec_10x10 | EEE-FK1H221P | Panasonic |
| C10 C14 | 2 | 22p | C_0603_1608Metric_Pad1.08x0.95mm_HandSolder | AC0603JRNPO9BN220 | Yageo |
| C15 C17 C18 | 3 | 22u | C_1210_3225Metric_Pad1.33x2.70mm_HandSolder | MCAST32MSB7226KPNA01 | TAIYO YUDEN |
| C12 | 1 | 33p | C_0603_1608Metric_Pad1.08x0.95mm_HandSolder | AC0603JRNPO9BN330 | Yageo |
| C2 C4 | 2 | 4.7u | C_1206_3216Metric_Pad1.33x1.80mm_HandSolder | C1206C475K5RACAUTO | KEMET |
| D1 | 1 | BAV99 | SOT-23 | BAV99Q-7-F | Diodes Inc |
| J1 | 1 | B4B XH A | JST_XH_B4B-XH-A_1x04_P2.50mm_Vertical | B4B XH A | JST |

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**Desk controller**

**Notes**

Sheet: Top BOM - fitted (1 of 3)
File: Desk_controller.kicad_pcb
Rev: 1.0.1 | Drwn: FLR
Mat: Concept | Rev'd: FLR
Date: 06/10/2026 | Rls'd: FLR
Dwg: NA | Size: A4
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# Top BOM - fitted (2 of 3)

| References | Qty | Value | Footprint | MPN | Manufacturer |
|---|---|---|---|---|---|
| J7 | 1 | B6B XH A | JST_XH_B6B-XH-A_1x06_P2.50mm_Vertical | B6B XH A | JST |
| J8 | 1 | DC barrel assy | Dell_barrel_assy | NA | NA |
| J6 | 1 | Dell DC jack | Dell DC jack | Barrel jack | Dell |
| J5 | 1 | USB_C_Receptacle_USB2.0_16P | USB_C_socket_16P_vertical | TFKH5-16L4CC-65 | ChengHaoRan |
| JP4 JP5 JP7 JP14 JP19 | 5 | 0R | R_1206_3216Metric_Pad1.30x1.75mm_HandSolder | Hand cut wire | NA |
| JP18 JP22 | 2 | 0R | R_Axial_DIN0204_L3.6mm_D1.6mm_P7.62mm_Horizontal | Hand cut wire | NA |
| JP6 JP20 | 2 | 0R | R_0603_1608Metric_Pad0.98x0.95mm_HandSolder | RC0603JR-130RP | Yageo |
| JP26 | 1 | 0R | Wire_6.853x8.890mm | Hand cut wire | NA |
| JP9 | 1 | 0R | R_Axial_DIN0204_L3.6mm_D1.6mm_P5.08mm_Horizontal | Hand cut wire | NA |
| L1 | 1 | 10u | L_Bourns_SRP6540 | SRP6540-100M | Bourns |
| R1 R2 R7 R8 R9 | 5 | 10k | R_0603_1608Metric_Pad0.98x0.95mm_HandSolder | AC0603FR-0710KL | YAGEO |
| R6 | 1 | 23.2k | R_0603_1608Metric_Pad0.98x0.95mm_HandSolder | AC0603FR-0723K2L | YAGEO |

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**Desk controller**

**Notes**

Sheet: Top BOM - fitted (2 of 3)
File: Desk_controller.kicad_pcb
Rev: 1.0.1 | Drwn: FLR
Mat: Concept | Rev'd: FLR
Date: 06/10/2026 | Rls'd: FLR
Dwg: NA | Size: A4
Generated with KiCad | Page: 5/7

# Top BOM - fitted (3 of 3)

| References | Qty | Value | Footprint | MPN | Manufacturer |
|---|---|---|---|---|---|
| R4 R5 R10 R12 R13 | 5 | 2k2 | R_0603_1608Metric_Pad0.98x0.95mm_HandSolder | AC0603FR-072K2L | YAGEO |
| R14 R15 | 2 | 5k1 | R_0603_1608Metric_Pad0.98x0.95mm_HandSolder | AC0603FR-075K1L | YAGEO |
| R3 | 1 | 64.9k | R_0603_1608Metric_Pad0.98x0.95mm_HandSolder | AC0603FR-0764K9L | YAGEO |
| R11 | 1 | 820 | R_0603_1608Metric_Pad0.98x0.95mm_HandSolder | AC0603FR-07820RL | YAGEO |
| U1 | 1 | LMR380x0 | Texas_HSOP-8-1EP_3.9x4.9mm_P1.27mm_ThermalVias_w_step | LMR38020FDDAR | Texas instruments |
| U2 | 1 | STM32G431K8T | LQFP-32_7x7mm_P0.8mm | STM32G431K8T6 | ST |
| U4 | 1 | USBLC6-2SC6 | SOT-23-6 | USBLC6-2SC6 | ST |
| U3 | 1 | VNH7070 | JEITA_SOIC-16_3.9x9.9mm_P1.27mm | VNH7070 | ST |
| Y1 | 1 | 20 MHz | Crystal_SMD_3225-4Pin_3.2x2.5mm_HandSoldering | CX3225SB20000H0FLJCC | KYOCERA AVX |

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**Desk controller**

**Notes**

Sheet: Top BOM - fitted (3 of 3)
File: Desk_controller.kicad_pcb
Rev: 1.0.1 | Drwn: FLR
Mat: Concept | Rev'd: FLR
Date: 06/10/2026 | Rls'd: FLR
Dwg: NA | Size: A4
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# Bottom BOM - fitted (1 of 1)

| References | Qty | Value | Footprint | MPN | Manufacturer |
|---|---|---|---|---|---|
| J2 | 1 | 01x04 socket | PinSocket_1x04_P2.54mm_Vertical | Generic | Generic |
| J3 | 1 | 01x14 socket | PinSocket_1x14_P2.54mm_Vertical | Generic | Generic |
| J4 | 1 | HX JN1.27-2x5 WZ H4.9 | JN1.27-2x5 WZ H4.9 | HX JN1.27-2x5 WZ H4.9 | Hanxia |
| JP23 | 1 | 0R | Wire_9.603mm | Hand cut wire | NA |
| JP24 | 1 | 0R | Wire_6.310mm | Hand cut wire | NA |
| JP25 | 1 | 0R | Wire_9.532mm | Hand cut wire | NA |
| TFT1 | 1 | TFT_2.4in_320x240 | Display_240x320_ILI9341_2.4in_no_pads | ILI9341_320x240 | TENSTAR ROBOT |

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**Desk controller**

**Notes**

Sheet: Bottom BOM - fitted (1 of 1)
File: Desk_controller.kicad_pcb
Rev: 1.0.1 | Drwn: FLR
Mat: Concept | Rev'd: FLR
Date: 06/10/2026 | Rls'd: FLR
Dwg: NA | Size: A4
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